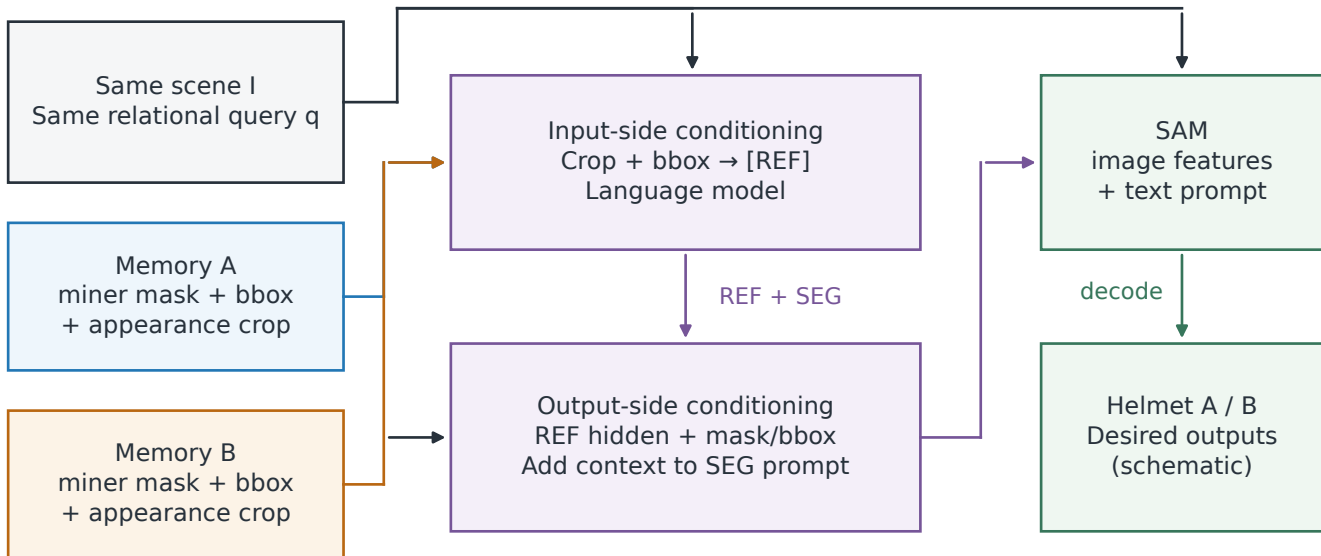


Supplied identity-localized entity memory

Same-condition appearance; externally supplied, fixed miner geometry
image I



Switch A ↔ B; hold *I*, *q* fixed

Shared CMA weights; one identity condition per trial

Training only: paired identity rank

$$S_{ij} = \text{softIoU}(p_i, Y_j)$$

$$[\mu + \max_{j \neq i} S_{ij} - S_{ii}]_+$$

Alongside text, mask and REF reconstruction losses

Offline evaluation: saved binary masks

JAA	JAB
JBA	JBB

Fidelity: correct > wrong

CMSA: both correct + IoU ≥ 0.5

IER: stronger wrong IoU ≥ 0.5